

PROVISIONAL APPLICATION FOR PATENT

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TITLE OF THE INVENTION

Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation

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CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to co-pending provisional patent applications filed on February 2, 2026: (1) "Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems"; and (2) "Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers."

FIELD OF THE INVENTION

[0002] The present invention relates to spiking neural network architectures, and more specifically to a method and system for enabling a spiking neural network to observe its own prior computational output through a configurable recursive feedback pathway with dynamically adjustable gain, creating a tunable spectrum of self-referential processing.

BACKGROUND OF THE INVENTION

[0003] Current neural network architectures, including both artificial neural networks (ANNs) and spiking neural networks (SNNs), lack a mechanism for explicit self-observation. While recurrent neural networks (RNNs), including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures, feed hidden state information back into the network, this feedback serves computational purposes. It enables the network to maintain memory and context for sequential processing.

[0004] No existing architecture provides a dedicated self-observation channel separate from the computational feedback, a continuously tunable awareness parameter that controls the degree of self-referential processing, dynamic external modulation of the self-observation gain during operation, or measurable correlation between the self-observation signal and behavioral state.

[0005] Recurrent neural networks such as LSTMs, GRUs, and Transformers use hidden state feedback for computation. The feedback is an implementation detail of the learning algorithm, not an explicit self-observation mechanism.

[0006] Reservoir computing and echo state networks use recurrent dynamics in a fixed random network as a computational substrate. The recurrence serves temporal processing, not self-observation. There is no configurable gain on the self-referential signal.

[0007] Metacognitive AI systems reason about their own confidence or uncertainty but do so through separate metacognitive modules, not through recursive observation of their own neural dynamics. [0007a] Global Workspace Theory implementations broadcast information across modules but do not feed a module's own output back to itself with configurable gain. The broadcast is inter-module, not intra-module self-observation.

[0008] There exists no prior system that records the aggregate output of a neural network at each timestep, feeds that recorded output back as explicit input to the same network at the next timestep, scales this feedback by a configurable coefficient that can be adjusted during operation, creates a measurable tunable spectrum from zero self-observation to full self-dominance, and enables external cognitive layers to dynamically control the degree of self-awareness.

SUMMARY OF THE INVENTION

[0009] The present invention provides a method and system for recursive self-observation in spiking neural networks comprising: (1) recording the aggregate output, specifically the mean membrane potential, of a spiking neural network at each timestep t ; (2) feeding back this recorded output as additional input to the same network at timestep $t+1$; (3) scaling the feedback by a configurable reflection coefficient ranging from 0.0 to 1.0; (4) dynamic modulation of said reflection coefficient by an external cognitive control layer or internal energy-aware regulation mechanism; and (5) measurable distinction between the self-observation signal and noise, with demonstrable correlation to system behavioral state.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a block diagram of the self-observation feedback loop showing external input 10, summing junction 12, spiking neural network 14, spike output 16, aggregate output 18, delay element 20, reflection scaling block 22, and self-observation feedback path 24.

[0011] FIG. 2 is a spectrum diagram showing the reflection coefficient spectrum bar 26 across its full range from 0.0 to 1.0, including no self-observation region 28, minimal awareness region 30, balanced operational region 32, self-dominated region 34, self-absorption region 36, and current processing point 38.

[0012] FIG. 3 is a block diagram showing the dual modulation sources that dynamically control the reflection coefficient value, including spiking neural network 14, reflection coefficient computation block 40, external cognitive control 42, external modulation examples 44, energy state monitor 46, energy rules box 48, output arrow 50, source priority box 52, and output to SNN 54.

[0013] FIG. 4 is a flow diagram of the self-referential learning loop showing spiking neural network 14, STDP weight update block 56, updated weight matrix block 58, self-observation block 60, key insight box 62, and validation results box 64.

[0014] FIG. 5 is a control diagram showing energy-aware self-observation regulation including energy state bar 66, modulation mapping row 68, reflection coefficient scale 70, behavioral effects row 72, and optional extension zone 74.

[0015] FIG. 6 is an end-to-end signal flow diagram showing spiking neural network 14, external cognitive control 42, self-observation block 60, environmental input block 76, motor actuation block 78, and energy harvesting block 80, illustrating how the self-observation mechanism integrates within the complete self-sustaining cognitive loop and distinguishing the self-observation feedback pathway from the energy feedback and synaptic computation pathways.

DETAILED DESCRIPTION OF THE INVENTION

[0016] Referring to FIG. 2, the reflection coefficient spectrum bar 26 creates a tunable self-awareness spectrum. At 0.0 the network occupies the no self-observation region 28 and processes only external input. At 0.2 the network operates in the minimal awareness region 30 with subtle influence on neural dynamics. In the balanced operational region 32, the system runs at its default configuration. At 0.5 to 0.7 the network enters the self-dominated region 34 and is significantly influenced by its own prior state. At 0.7 to 1.0 the network approaches the self-absorption region 36 and is dominated by its own prior output. The current processing point 38 indicates the active reflection coefficient value on the spectrum at a given moment. Core Mechanism

[0017] Referring to FIG. 1, at each timestep t the spiking neural network 14 receives external input 10 through a summing junction 12 and processes it through leaky integrate-and-fire dynamics. For each neuron i , the membrane potential $V_i(t)$ equals $V_i(t-1)$ multiplied by the quantity $1 - \text{leak_factor}$, plus $I_{\text{ext}}(t)$ multiplied by input_scale multiplied by the Kronecker delta of i and 0, plus $I_{\text{reflection}}(t)$, plus the sum over j of w_{ij} multiplied by $\text{spike}_j(t)$. The external input 10 enters the spiking neural network 14 at neuron 0 via the summing junction 12, and w_{ij} represents the synaptic weight from neuron j to neuron i . The resulting spike output 16 of the network constitutes the per-neuron firing pattern at this timestep. Recording Aggregate Output

[0018] After spike generation and propagation, the network's aggregate output 18 is computed and held in a delay element 20 as previous_output equals the mean of $V_i(t)$ for all neurons i . This single scalar value, derived from the spike output 16 via mean-pooling of the membrane potential vector, represents the network's overall activation state and constitutes a compressed representation of the full neural state. The delay element 20 makes the aggregate output 18 available to the next timestep as the source of the

self-observation feedback described in the following paragraph. Self-Observation Feedback

[0019] At timestep $t+1$, the recorded aggregate output 18 held in the delay element 20 is fed back as additional input along the self-observation feedback path 24, passing through the reflection scaling block 22 whose gain is the configurable reflection coefficient. The reflection input $I_{\text{reflection}}(t+1)$ equals the previous_output multiplied by the reflection coefficient. This feedback is added to the external input 10 at the input neuron of the spiking neural network 14 via the summing junction 12 such that the adjusted input equals $I_{\text{ext}}(t+1)$ multiplied by input_scale plus previous_output multiplied by reflection_coeff, and $V_0(t+1)$ is incremented by the adjusted input. Dynamic Modulation

[0020] Referring to FIG. 3, the reflection coefficient is not fixed but can be dynamically adjusted during operation through dual modulation sources arbitrated by a reflection coefficient computation block 40. The first source is external cognitive control 42, which provides an externally supplied modulation signal; representative external modulation examples 44 include goal-directed setpoints, attention-allocation decisions, and task-switch commands. For external modulation through cognitive control 42, the reflection coefficient equals base_reflection plus modulation multiplied by sensitivity, where base_reflection is 0.2 as the default baseline, modulation is an external cognitive signal ranging from negative 1.0 to positive 1.0, and sensitivity is 0.1 as the modulation gain. At modulation of negative 1.0 the reflection coefficient is 0.1 representing minimal self-observation. At modulation of 0.0 the reflection coefficient is 0.2 representing baseline. At modulation of positive 1.0 the reflection coefficient is 0.3 representing enhanced self-observation. The computed reflection coefficient is emitted along an output arrow 50 as output to SNN 54, where it controls the gain of the reflection scaling block 22 described in paragraph 0019.

[0021] The second modulation source is internal energy-aware modulation. An energy state monitor 46 reads the current energy reserve and applies the thresholds defined by an energy rules box 48 as follows. Referring to FIG. 5, the modulation mapping row 68 maps positions on the energy state bar 66 to entries on the reflection coefficient scale 70, producing the outcomes listed in the behavioral effects row 72: when energy is below 15 milliwatt-hours the modulation is negative 0.4 to conserve and reduce self-observation; when energy is below 30 milliwatt-hours the modulation is negative 0.1 for cautious operation; when energy exceeds 80 milliwatt-hours the modulation is positive 0.3 to increase self-observation for exploration and learning; otherwise the modulation is 0.0 for neutral operation. An optional extension zone 74 on the energy state bar 66 is reserved for additional plateaus that may be added in future refinements without altering existing thresholds. A source priority box 52 inside the reflection coefficient computation block 40 selects between the external cognitive control 42 signal and the internal energy-aware signal; typical implementations prefer the external signal when present and fall back to the internal signal in its absence. This creates a biologically plausible dynamic wherein when resources are scarce, the system reduces self-referential processing to focus on survival, and when resources are abundant, it increases self-observation for learning and adaptation.

Implementation

[0022] The self-observation mechanism is implemented in the neural network step method. The membrane potential is first decayed by the leak factor. The adjusted input is computed by scaling the external input signal by the input scale and, if a previous output exists and the reflection coefficient is greater than zero, adding the previous output multiplied by the reflection coefficient. This adjusted input is applied to the first neuron. After spike detection, propagation, and optional STDP weight update, the network's aggregate output is recorded as the mean of all membrane potentials. This recorded value persists across timesteps and is fed back at configurable gain.

[0023] The previous_output field is the self-observation state. It is architecturally distinct from the synaptic weights which serve computation, and from the spike history which serves recording. Interaction with STDP Learning

[0024] Referring to FIG. 4, when the spiking neural network 14 has Spike-Timing Dependent Plasticity enabled, the self-observation feedback creates a novel interaction represented by the self-observation block 60. The reflection signal modulates the input neuron's firing pattern. This changes the timing relationships between the input neuron and downstream neurons. The STDP weight update block 56 adjusts synaptic weights based on these timing relationships, producing an updated weight matrix block 58. The adjusted weights change the network's response to subsequent reflection signals. As summarized in the key insight box 62, this creates a self-referential learning loop wherein the network learns about its own dynamics through the interaction of self-observation and synaptic plasticity. The validation results box 64 records the F8.1-F8.3 falsifiable outcomes (weight-entropy decrease, mutual information gain, and weight-topology convergence) that empirically confirm the loop's operation. Measurable Properties

[0025] The self-observation signal can be distinguished from noise through several measurable properties. The reflection signal has temporal autocorrelation structure that matches the input-driven dynamics, unlike white noise which has zero autocorrelation at all lags greater than zero. The reflection state correlates with system behavioral state including spike pattern type, energy level, and temperature in ways that random noise does not. Changing the reflection coefficient produces measurable changes in mean firing rate, spike pattern classification, weight entropy when STDP is enabled, and energy trajectory.

EXPERIMENTAL VALIDATION

[0026] Phase 7 Control Baseline with Claims A through E demonstrates that the system with reflection coefficient of 0.2 passes all five falsifiable claims, including input-output gain at Claim D with correlation at least 0.2. Setting the reflection coefficient to 0.0 changes the system dynamics measurably.

[0027] Phase 8 STDP validation with Claims F8.1 through F8.3 demonstrates that the STDP network with self-observation feedback achieves 6.52 times higher mutual information between input and output compared to a frozen-weight control, as validated by Claim F8.2. The self-observation pathway contributes to this information gain by providing the network with temporal context about its own prior state.

[0028] The MCP operational path demonstrates that the cognitive control layer dynamically adjusts the reflection coefficient based on pattern type, energy state, and attention allocation. The system's behavior measurably changes with different modulation values. At modulation of negative 0.4 for conservation there is reduced spike activity and lower energy consumption. At modulation of 0.0 for neutral operation there is baseline dynamics. At modulation of positive 0.3 for exploration there is increased spike activity and higher energy consumption. End-to-End Signal Flow

[0029] Referring to FIG. 6, the complete end-to-end signal flow is shown for a system incorporating recursive self-observation. An environmental input block 76 representing sensors produces input signals that enter the spiking neural network 14. The network encodes these signals as spike patterns while simultaneously receiving self-observation feedback from its own prior aggregate output via the self-observation block 60, scaled by the reflection coefficient. The external cognitive control 42 stage adjusts the reflection coefficient dynamically based on external cognitive commands or internal energy state. The modulated neural output drives a motor actuation block 78, which generates movement. Energy harvested by an energy harvesting block 80 from that movement

powers the network. The self-observation block 60 represents a feedback pathway architecturally distinct from both the energy feedback pathway emanating from the energy harvesting block 80 and the synaptic computation pathway internal to the spiking neural network 14. This diagram illustrates how the self-observation mechanism integrates within the broader self-sustaining cognitive loop described in the co-pending application for the Neural-Motor Energy Harvesting Loop. Scope of Claims

[0030] No claim is made herein regarding phenomenological consciousness, subjective experience, qualia, or sentience. The term "self-observation" as used throughout this specification refers exclusively to the engineering mechanism of feeding a neural network's aggregate output back as measurable input at configurable gain. The term "self-awareness" refers to the degree of self-referential signal magnitude relative to external input magnitude, as quantified by the reflection coefficient. All claims pertain to measurable, falsifiable properties of signal feedback, gain modulation, and neural dynamics. The system is a deterministic engineering artifact whose operation is fully characterized by the mathematical equations and validation thresholds described herein.

CLAIMS

[0031] What is claimed is:

1. A method for enabling recursive self-observation in a spiking neural network comprising:

- (a) processing input signals through a spiking neural network comprising a plurality of leaky integrate-and-fire neurons to produce membrane potentials and spike events;
- (b) computing an aggregate output value from said membrane potentials at timestep t , said aggregate output representing a compressed state observation of the network's overall activation;
- (c) storing said aggregate output value as a self-observation state;
- (d) at timestep $t+1$, feeding said stored self-observation state back as additional input to said spiking neural network, scaled by a reflection coefficient;
- (e) dynamically adjusting said reflection coefficient during operation based on at least one of: an external cognitive control signal, an internal energy-aware regulation signal, or a combination thereof; wherein said self-observation feedback creates a tunable spectrum of self-referential processing ranging from zero self-observation when the reflection coefficient equals zero to maximum self-observation when the reflection coefficient equals one, and wherein said self-observation signal is measurably distinct from noise and correlates with system behavioral state.

2. A spiking neural network system with configurable self-observation comprising:

- (a) a spiking neural network processor having a plurality of neurons, each having a membrane potential, a firing threshold, and synaptic connections;
- (b) an output aggregation module that computes a scalar aggregate output from said membrane potentials at each timestep;
- (c) a self-observation state register that stores said aggregate output between timesteps;
- (d) a reflection coefficient register that stores a configurable scalar value between 0.0 and 1.0;
- (e) a feedback injection module that multiplies said stored self-observation state by said reflection coefficient and adds the result to the input of said spiking neural network at the subsequent timestep;
- (f) a modulation interface that allows dynamic adjustment of said reflection coefficient during operation; wherein said system provides a continuously tunable degree of self-referential processing without architectural changes to the underlying spiking neural network.

3. The method of claim 1 wherein said aggregate output value is computed as the arithmetic mean of membrane potentials across all neurons in the network, being one divided by N multiplied by the sum over i of $V_i(t)$, where N is the number of neurons and $V_i(t)$ is the membrane potential of neuron i at timestep t .

4. The method of claim 1 wherein said external cognitive control signal is provided by a separate reasoning system through a defined protocol interface, said reasoning system analyzing the neural network's state including spike pattern type, energy level, and temperature and determining an appropriate modulation value in the range of negative 1.0 to positive 1.0.

5. The method of claim 1 wherein said internal energy-aware regulation signal is computed from the system's current energy storage level according to a multi-threshold rule: when energy is below a critical threshold the modulation is negative 0.4 for maximal conservation; when energy is below a low threshold the modulation is negative 0.1 for cautious operation; when energy exceeds a high threshold the modulation is positive 0.3 for exploration; and otherwise the modulation is 0.0 for neutral operation.

6. The method of claim 1 wherein said reflection coefficient is computed as $\text{reflection_coeff} = \text{base_reflection} + \text{modulation} \times \text{sensitivity}$, where base_reflection is a configurable baseline value defaulting to 0.2, modulation is a control signal in the range of negative 1.0 to positive 1.0, and sensitivity is a configurable gain defaulting to 0.1.

7. The method of claim 1 further comprising applying spike-timing dependent plasticity to the synaptic weights of said spiking neural network simultaneously with said self-observation feedback, wherein the self-observation signal modulates the firing pattern of at least one neuron, STDP adjusts synaptic weights based on temporal correlations between neuron firings, and the combination creates a self-referential learning loop wherein the network learns about its own dynamics through the interaction of self-observation and synaptic plasticity.

8. The method of claim 1 wherein the self-observation feedback is injected at a single designated input neuron of said spiking neural network, combined with external environmental input such that V_{input} at $t+1$ is incremented by the sum of external_input multiplied by input_scale and stored_output multiplied by reflection_coeff .

9. The system of claim 2 wherein said modulation interface supports at least two modulation sources operating simultaneously: (a) an external cognitive control layer that provides goal-directed modulation based on reasoning about the network's state; and (b) an internal energy-aware regulator that provides survival-oriented modulation based on the system's energy reserves; and wherein the final reflection coefficient is determined by a combination of both sources.

10. The method of claim 1 used in combination with a self-sustaining neural-motor energy harvesting loop, wherein the spiking neural network's motor output generates energy through piezoelectric and thermoelectric harvesting, the harvested energy powers the neural network's continued operation, and the self-observation feedback enables the system to modulate its own behavior for energy homeostasis, such that the degree of self-observation is adapted based on the system's energy state.

ABSTRACT OF THE DISCLOSURE

A method and system for configurable recursive self-observation in spiking neural networks. At each timestep, the aggregate output, being the mean membrane potential, is recorded and fed back as additional input at the subsequent timestep, scaled by a reflection coefficient ranging from 0.0 (no self-observation) to 1.0 (maximum self-observation). The reflection coefficient is dynamically adjustable through an external cognitive control layer providing goal-directed modulation, an internal energy-aware regulation mechanism, or both. This creates a tunable self-awareness parameter enabling a continuous spectrum of self-referential processing without architectural changes. When combined with spike-timing dependent plasticity, the self-observation creates a self-referential learning loop wherein the network learns about its own dynamics. Validated through falsifiable experiments demonstrating measurable behavioral changes across the reflection coefficient spectrum and 6.52 times higher mutual information compared to frozen-weight controls. (FIG. 1)

REFERENCE TO SOURCE CODE

The complete implementation of this invention is available at the following repository: github.com/DarkWinD90/Consciousness_Env, validated state tagged as v3.0.0-phase10-multimodal at commit 9e2c333. Key implementation file is `core/base_snn.py`. Dynamic modulation is implemented in `mcp/consciousness_mcp_server.py`.